⚡ Zero45 – Quick reference

As a team you plan Germany's electricity up to 2045 – in 4 stages. Each stage: up to 5 points for each of the three goals, at most 60 in total. **Around 40 points is a very good result.**

1 · How a stage works

1. **Build:** spend build tokens (€10 bn each) on new plants; keep or retire old power plants (retiring is final).
2. **Influence:** use influence tokens for measures (each once per game).
3. **Test:** simulator (3× per stage, normal weather, no chance) – shows the change since the last test.
4. **Mark as ready** – can be changed until the game master closes the phase. A notice appears 2 minutes before the end.
5. **Simulate the year:** the same chance for everyone (weather, demand, disaster) → points and a 💡 tip for the next stage.

2 · The three goals

- ⚡ **Security of supply:** hours in which power is missing.
- 🌿 **Climate protection:** share of CO_2-free power *and* CO_2 emissions in Germany – **the worse value counts**. Exports count towards the share, imports do not.
- 💶 **Affordability:** generation and storage costs per MWh (excl. grids, taxes).

3 · What there is

- **Wind & sun:** cheap, but weather-dependent. Old plants go offline step by step.
- **Batteries, biomass flex:** for hours, not for weeks.
- **New gas turbines:** firm capacity; convertible to H_2 with a hydrogen card (then CO_2-free).
- **Hydrogen:** own electrolysis + cavern (demonstration programme) or imports (hydrogen grid).
- **Credit:** up to 4 extra tokens per stage, +20 % costs.
- **Protest:** every token can trigger protest → technology blocked for one stage; in stage 4 −2 points. Build broadly!

4 · Influence cards (2 tokens at the start, more for 2 build tokens each)

Policy & society

- **Acceptance & participation:** Halves the risk of public backlash – from this stage.
- **Accelerated innovation:** Double cost reduction from your own construction projects – from this stage.
- **Climate resilience:** Halves the impact of disasters (including dark doldrums, heat) – from this stage.

Technology & infrastructure

- **Hydrogen demonstration programme:** Your own green hydrogen: electrolysis, storage and conversion of gas turbines – from the next stage.
- **Hydrogen grid & imports:** Imported hydrogen for converted gas turbines, plus storage and conversion (no electrolysis) – from the next stage.
- **European grid expansion:** +10 GW of interconnectors to other countries for imports and exports – from the next stage.
- **Flexible demand:** Electric cars, heat pumps and industry adapt their consumption – from the next stage.

5 · The stages

	Now–2030	2031–35	2036–40	2041–45
Demand	599 TWh	809 TWh	928 TWh	980 TWh
Peak load	106 GW	143 GW	163 GW	166 GW
Coal max.	17 GW	6 GW	0 GW	0 GW
Build tokens	18	18	17	17

6 · Point thresholds

Points	5	4	3	2	1
Gap hours ≤	3	10	25	50	100
CO_2-free ≥ % (Now–2030)	80	75	70	65	60
CO_2-free ≥ % (2031–35)	88	84	80	76	72
CO_2-free ≥ % (2036–40)	93	90	87	84	80
CO_2-free ≥ % (2041–45)	97	95	92	89	85
CO_2 ≤ Mt (Now–2030)	70	80	90	105	120
CO_2 ≤ Mt (2031–35)	45	55	65	80	95
CO_2 ≤ Mt (2036–40)	25	35	50	65	80
CO_2 ≤ Mt (2041–45)	8	20	35	50	65
Costs ≤ €/MWh (Now–2030)	58	63	68	73	79
Costs ≤ €/MWh (2031–35)	64	69	74	80	86
Costs ≤ €/MWh (2036–40)	63	68	74	80	87
Costs ≤ €/MWh (2041–45)	65	70	76	83	91

> **Firm capacity** (right side of the Build tab): dispatchable power plants compared with the peak load. Two traffic lights: **+ storage & flex** (green from +10 %) and **+ interconnectors** (green from +30 %). Hydrogen turbines only count as far as the hydrogen lasts for 2 weeks.

7 · Tips

1. Test first, then mark as ready – the simulator knows no chance, plan a reserve.
2. Use up your build tokens, leftovers expire. Building early pays off in all later stages.
3. Stage 2 is the big jump: much more demand, hardly any coal left.
4. Think about hydrogen for 2040/45 – the cards only take effect one stage later.
5. The CO_2 target falls every stage: replace old gas power plants in time.